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Periodic poling with a micrometer-range period in thin-film lithium niobate on insulator

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We present our experimental results on electric field poling of X-cut thin-film lithium niobate with poling periods of 1–4 µm. We used polarization contrast microscopy combined with digital image processing to study the impact of the experimental parameters of the poling process such as electric field strength, poling pulse duration, and layout of metal electrodes on the domain growth and quality of resulting poling patterns. High-quality results with excellent homogeneity of the domain pattern on large areas [i.e., 16 µm along the domain (z axis) and millimeter long domain gratings along the y axis with a period down to 1 µm] allow for the integration of numerous devices and structures.

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1. INTRODUCTION

Lithium niobate (LN) is a favorable material for many applications in photonics due to its large transparency range, high nonlinearity, and the ability for electro-optic as well as acousto-optic modulation of its properties [1]. In recent years, thin films of high-quality crystalline lithium niobate with only a few hundreds of nm thickness became available. This LN on insulator (LNOI) is considered as a very promising platform for nanoscale photonic integrated circuits (PICs) [2], since waveguides with a small cross section and high refractive index difference can be realized. Such PICs can confine light to very small cross sections, enabling smaller and more compact device footprints compared to diffusion-based waveguide technologies [3]. However, one of the main applications of LN, which particularly profits from the strong field confinement of nanoscale PICs in LN, is nonlinear frequency conversion by parametric three-wave mixing, such as second-harmonic generation (SHG) [4–10], difference frequency generation (DFG) [11,12], and spontaneous parametric down-conversion (PDC) [13]. However, the efficiency of these nonlinear processes can only be ensured by phase matching the interacting waves. In LN PICs, the most-common technique to realize phase matching is quasi-phase matching (QPM) [1,14]. In this method, the crystal structure of LN is periodically inverted along the crystalline z axis. The resulting periodically poled LN (PPLN) thus in effect provides an additional wavevector due to the periodic spatial modulation of nonlinear susceptibility to match the phase velocities of the interacting waves.

Different methods to fabricate PPLN have been introduced [2,10,15–17]; however, electric field poling (EFP) is the most used [4,18–22]. In this method, an electric field larger than the coercive field strength of LN is applied along the crystalline z axis to permanently invert the spontaneous polarization of LN. For bulk Z-cut LN wafers up to several mm in thickness, the domain inversion with EFP is done using suitably patterned electrodes on the $+z$ and $-z$ faces of the wafer. Applying a voltage on these electrodes produces a field through the LN crystal and leads to local domain inversions [1,9,18,19,23–26].

However, EFP through the entire thickness of a Z-cut LNOI wafer is challenging due to the intermediate insulating layer [4] blocking the domain growth. Consequently, EFP has been used in most demonstrations to date along the surface of a X- or Y-cut LN thin film only [6–8,20–22]. For surface EFP, both positive and negative electrodes are deposited on the same surface of thin-film LN with the crystalline z axis in the plane of the film along the direction of the electric field, as shown schematically in Fig. 1(a). By applying an electric signal to the electrodes, the domains grow from the positive electrode side to the negative electrode side in a volume close to the surface. Waveguides can then be structured perpendicular to the poled domains in the thin film.

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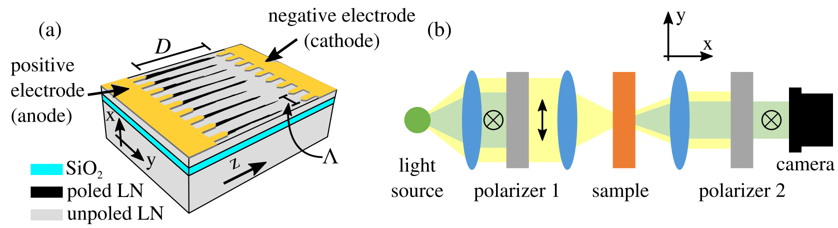

Fig. 1. (a) Schematic of X-cut LNOI with metal electrodes on top with a period of Λ and a distance between electrodes of D. Domains, shown in black, grow from the positive electrode to the negative electrode. (b) Setup to inspect poling patterns using PCM. Polarizers 1 and 2 are adjusted perpendicular to each other.

Depending on the propagation constants of the waves interacting in the frequency-mixing processes, different poling periods are required to achieve phase matching. For example, for second-harmonic generation (SHG) with co-propagating waves with a fundamental wavelength of 1550 nm in a LNOI waveguide, the required poling period is on the order of a few micrometers [7,8]. However, for backward SHG with counter-propagating fundamental and second-harmonic waves, poling periods smaller than 1 μm are needed in bulk [9] and waveguide geometries [27] due to the large wavevector mismatch.

Regardless of the targeted period, optimal nonlinear conversion efficiency is based on a good homogeneity of the PPLN throughout the poled region. This is challenging, since domain growth, triggered by ms-long electric voltage pulses, is a fast and dynamic process starting spontaneously from a few nucleation sites in the LN crystal. Several works have derived optimal poling conditions for poling periods of several μm and larger in LNOI [6–8,20,21,28]. Poling of LN with a large period is more flexible and different methods such as applying a long signal or several short signals can be used to get high-quality results without merging of domains. However, for small domains in PPLN patterns with small periods, a precise control and understanding of the poling process is needed to avoid effects like the merging of neighboring domains. This requires an understanding of the mechanism of domain growth during the poling process.

In this work, we fabricate a periodically poled LNOI with periods Λ between 1 and 4 μm. Using a rapid nondestructive characterization technique, we statistically analyze realized poling patterns for different fabrication parameters like the period Λ, the width D of the poled region, and the parameters of the voltage pulse used for EFP. Our systematic approach enabled us to better understand the mechanism of domain growth during the poling process and the influence of changing the mentioned parameters on it, which could be used for optimization of the poling process to achieve high-quality poling results in a wide area between the electrodes. In the following, first we explain our experimental realization and analysis method and then we present the results of our experiments.

2. EXPERIMENT AND ANALYSIS METHOD

For our poling experiments we use X-cut congruent LNOI as the substrate. The thickness of the film is 600 nm and it is on a 2 μm thick layer of SiO_2 and a 500 μm thick layer of LN for handling. A scheme of the sample patterned for poling is shown in Fig. 1(a). The poling electrode pattern is created on the surface of LNOI by electron beam lithography and a lift-off process. The electrodes consist of a 170 nm thick Cr layer, covered by a 30 nm Au layer for protection. Using Cr as a material for the electrodes increases the nucleation site density, which serves as the starting point of the poling process [23]. The patterned areas we used in our experiments have different lengths in the y direction from 600 μm to 4 mm. The electrodes are structured along the area to be poled where, for every domain that should be inverted, one protruding finger is used on the electrode, as indicated in Fig. 1(a). Previous works have shown that the tip shape of the electrode fingers plays an important role in the quality of the results [21,22]. Here, we use electrodes with triangular tips and proper width for different periods according to [22]. The poling is performed by applying a short voltage pulse to these electrodes. During the poling process, we keep the sample submerged in insulating silicone oil (BLUESIL FLD 604V50) to prevent the electric breakdown of air [29]. The poling signal is created by a computer-controlled, high-voltage amplifier (2HVA24-BP1-F-SHV-5KV, UltraVolt, Inc.) and we use additional measurement circuits for high-precision voltage and current monitoring.

To optimize the poling process and to find the best parameters for high-quality poling of LNOI, we perform a multitude of experiments to systematically investigate the influence of different parameters on the poling. This requires a fast, reliable investigation method to characterize the domains. For PPLN, different characterization methods have been developed: etching of the poled sample in a solution that is a combination of HF and HNO_3 with subsequent imaging of the domains [18], piezo-response force microscopy (PFM) [30], or second-harmonic microscopy [28,31,32].

In our work, we use a simpler, yet accurate technique, that allows for fast processing of many samples. It is based on polarization-contrast microscopy (PCM) and sketched in Fig. 1(b). This method uses a wide-field microscope in the transmission mode, where the light exciting the sample is linearly polarized. The sample is imaged through a polarizer, which transmits only the polarization perpendicular to the incoming polarization. This noncontact method can be used to characterize PPLN due to the spontaneous birefringence within the poled domains, which mainly originates from material stress after the poling process [33–35]. Compared to PFM and chemical etching, this nondestructive method has a lower resolution; however, by using objectives with a high numerical aperture (NA), poled domains as small as 1 μm can be reliably investigated. For our investigation, PPLN is imaged using a microscope objective

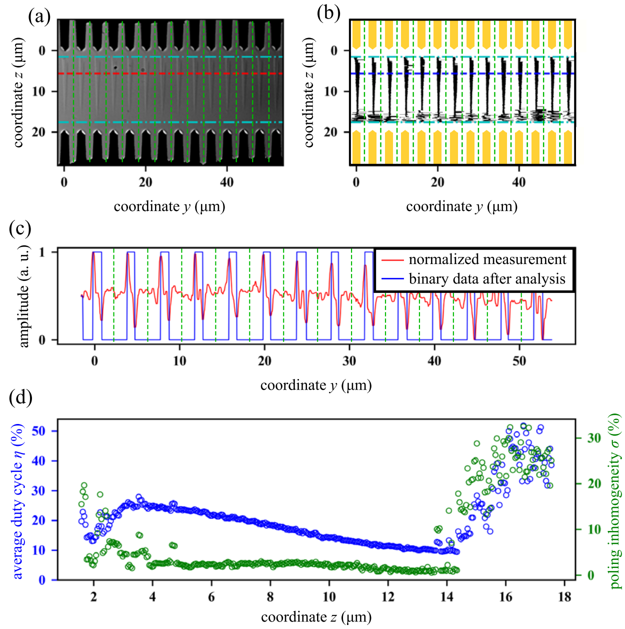

Fig. 2. Image processing method to analyze PCM images. (a) Microscope image of a poled region obtained using PCM. (b) Binary image of poled domains, where poled domains are black, unpoled regions are white, and electrodes are yellow. The analyzed region is between the horizontal dash-dotted lines in (a) and (b). (c) Normalized PCM measurement data (red) and binary data (blue) corresponding to the dashed lines in (a) and (b). For the binary data, the poled areas correspond to values of 1 and the unpoled areas are 0. Using the binary data, we calculate the average duty cycle η and standard deviation of duty cycle (poling inhomogeneity) σ. The dependence of η and σ along coordinate z in the analyzed region is shown in (d).

with a NA of 0.95 (EC Epiplan-Apochromat 150x, Carl Zeiss). With this measurement setup, the width of images along the y direction is around 50 μm [similar to Fig. 2(a)]. However, analyzing different PCM images from different areas of the sample shows a similar result independent of where on the sample we take the image.

To quantitatively analyze the images measured by PCM and to obtain statistics about the features of the domains, we developed a specific image processing algorithm, which is illustrated in Fig. 2. Figure 2(a) shows an example PCM image of poled LNOI. First, we divide the image area vertically [green dashed lines in Figs. 2(a)–2(c)] based on the period of the electrodes in a way that each pair of electrode fingers and the corresponding poled area are in one region. The subsequent analysis is restricted to a region of interest between the electrodes, which in this experiment had a width in the z direction of 16 μm and is bounded by the dash-dotted lines in Figs. 2(a) and 2(b). For each z coordinate, we then identified the poled regions along the y direction, resulting in binary images, as shown in Fig. 2(b), where the electrodes are yellow, the poled regions are black, and the unpoled LNOI is white. To identify the poled regions, we saw that the signal measured by PCM shows distinct maxima and minima at its boundaries [33,35]. This can be seen in Fig. 2(c), where we plot the normalized signal measured by PCM along the red dashed line in Fig. 2(a). Based on the found minimum and maximum positions, the poled regions can be identified. They are plotted as blue lines in Fig. 2(c). Poled areas are set to a value of +1 and unpoled areas are set to a value of 0. Performing this analysis for all z coordinates in the region of interest leads to the result shown in Fig. 2(b).

The properties of the realized poling patterns are characterized based on their duty cycle. The duty cycle of a domain is the ratio of the domain width to the intended poling period. Its value is individually calculated for all domains along the y direction for each z from the obtained domain boundaries. For each z position, we then calculate the average duty cycle η and its standard deviation σ, taking into account all domains along y. We refer to σ as poling inhomogeneity. Figure 2(d) shows these quantities for the datasets of Figs. 2(a) and 2(b) in dependence on the z position. In this example, we find that in the region between $z = 4$ μm and $z = 14$ μm, the average duty cycle decreases; i.e., the domains are getting narrower. In this region, the poling inhomogeneity is small, indicating a homogeneous poling pattern. In the regions closer to the electrodes, the poling inhomogeneity is much larger, corresponding to very inhomogeneous domain growth or the absence of poled domains. This can be attributed to the inhomogeneous edge fields formed near the tipped electrodes. This interpretation is confirmed by a processed PCM image shown in Fig. 2(b).

To evaluate the overall poling quality of the complete area between the electrodes and to compare different samples, we use all z-dependent data and calculate the total average duty cycle (η_{total}) and the total standard deviation of duty cycle (σ_{total}). We name σ_{total} as total poling inhomogeneity. These values are used in the following to study our poling experiments and to find the best parameters for poling LNOI.

3. RESULTS

In this study, we aim to fabricate structures with a distance between electrodes D of at least 20 μm. For $D = 20$ μm, the results of all experiments are analyzed in an area of 16 μm width along the z direction between the electrodes. The areas close to electrodes are excluded from the analysis as the poling quality is usually poor in these regions due to effects like back-switching [19]. The analyzed region of interest thus covers 80% of the whole area between the electrodes and is wide enough to fabricate several independent structures like waveguides and directional couplers in one region.

Our aim is to identify the poling conditions that result in the highest poling quality. Ideally, the domains should extend completely from the positive to the negative electrode and have the designed duty cycle, which in our experiments is 50%. Furthermore, the poled domain in each period along the y direction should have the same geometry; thus, the poling should be homogeneous across the complete area. For the whole poled area, these properties of the poled domains are captured by the total average duty cycle η_{total} and total poling inhomogeneity σ_{total}. A low σ_{total} demonstrates domains of similar width in all periods and for different z coordinate. Thus, these two parameters in the following will be used to optimize the poling signal for the best result.

The different electrical signals used for poling of LN [shown in Figs. 3(a), 3(c), and 3(f)], are freely adjustable sequences of linear functions. In this figure, U_p is the field strength, which is the ratio of the electric voltage we applied to the electrodes over the distance between the electrodes from tip to tip [D in Fig. 1(a)]. The time dependence of the main poling signal is shown in Fig. 3(a). It consists of a short rectangular pulse, shown with the blue shading, which has the highest voltage U_p and is responsible for the actual poling. Its duration is Δt_3 and it is complemented by additional voltage slopes before and after the main part, which are characterized by durations $\Delta t_{1,2}$ and $\Delta t_{4,5}$, respectively. Here, Δt_1 is the time to increase the voltage from zero to the voltage U_1 above the coercive field strength U_c. This is followed by a much faster increase of the voltage to U_p in the time Δt_2. After the main pulse, the voltage is reduced to U_1 in time Δt_4, before it is relatively slowly ramped to zero in Δt_5. The decrease of the voltage is slower than the increase to stabilize the domains after applying the poling field strength (U_p) [22]. A typical poling result after applying a signal with $\Delta t_1 = 30$ ms, $\Delta t_2 = 0.5$ ms, $\Delta t_3 = 3$ ms, $\Delta t_4 = 3$ ms, $\Delta t_5 = 300$ ms, $U_1 = 35$ V/μm, and $U_p = 54$ V/μm for a sample with period of 2 μm is shown in Fig. 3(b). Only small areas are poled in different places between the electrodes, continuous poled domains are not formed. This indicates a lack of nucleation sites in the sample, although better results were obtained for some samples. Hence, the main poling signal as described up to now is not sufficient for repeatable high-quality poling in our geometry. We do not get the consistent results reported in Z-cut LNOI using a single, but much longer, poling pulse [4] at a high temperature. The differences in the poling parameters between our experiments and [4] arise from the temperature difference and the different orientation of the poling direction with respect to the film interfaces in the LNOI wafer. There are two ways to increase the number of nucleation sites [22,36–38]. The first method uses an additional nucleation pulse just before the main poling pulse, as shown in Fig. 3(c) with the green shading, which has a field strength U_n slightly larger than U_p. For $U_n = 56$ V/μm, $U_p = 54$ V/μm, $\Delta t_n = 1$ ms, and $\Delta t_{3n} = 2$ ms, and the same timing for increasing and decreasing voltages as in the previous experiment, the PCM image of the resulting poling pattern is shown in Fig. 3(d). It shows an improvement in the domain growth, which now forms extended stripes between the electrodes. An alternative method to increase the nucleation site density is the use of pre-pulses before the actual poling pulse. The field strength needed for poling of thin-film LNOI is higher than the value needed for bulk LN [8]. For our poling configurations, poling does not happen for U_p smaller than 40 V/μm. We apply three pre-pulses, as shown in Fig. 3(e) before the main poling signal from Fig. 3(a). The field strength of the pre-pulses is $U_{pp} = 35$ V/μm, their time constants are $\Delta t_{pp} = 1$ ms, and $\Delta t_r = \Delta t_f = \Delta t_{wait} = 1$ ms. Applying these pulses before the main poling signal with the same parameters as described before leads to an improvement in the domain growth visible in the image of the domains in Fig. 3(f).

Both methods to increase the number of nucleation sites give promising and homogeneous results. Because the use of pre-pulses leads to a noticeable improvement in the homogeneity and domains growth from positive to negative electrodes and is also used in other reported experiments [37,38], for all further experiments we will apply three pre-pulses, as shown in Fig. 3(e). In the following, we adjust two main parameters of the poling signal, the poling field strength U_p, and the poling duration Δt_3 for different poling periods Λ and distances between electrodes D to maximize the poling quality. All other parameters of the poling pulses will be constant throughout these experiments, with the same values as mentioned earlier.

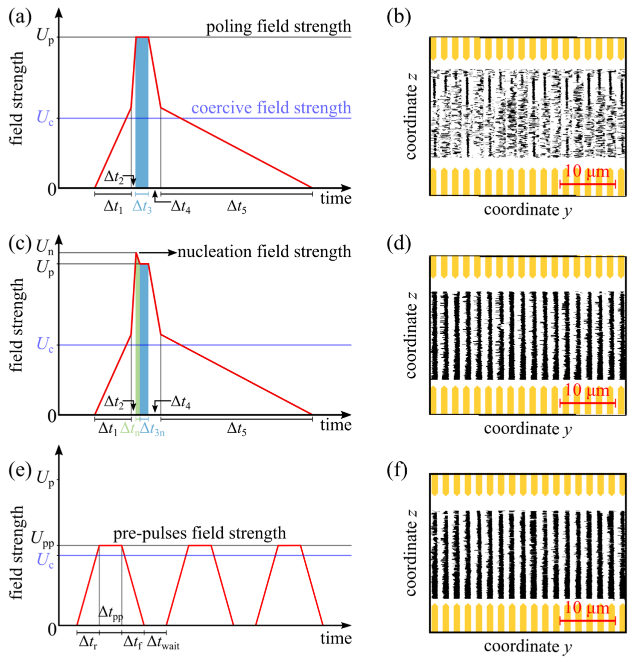

Fig. 3. (a) Time dependence of standard poling signal. (b) PCM image of a poling experiment resulting from signal (a). (c) Poling signal with additional nucleation pulse. (d) Result of poling experiment by applying the signal shown in (c). (e) Time dependence of pre-pulses, which are applied before the main poling signal to increase the number of nucleation sites. (f) Result of a poling experiment with applied pre-pulses, as shown in (e), and subsequent standard poling pulse, as shown in (a).

A. Experimental Results

First, we study the effect of changing the poling field strength U_p on the poling quality. For these experiments, we use a poling duration of $\Delta t_3 = 3$ ms and a distance between the electrodes of $D = 20$ μm. The impact of changing U_p on the total poling inhomogeneity σ_{total} for different poling periods is shown in Fig. 4(a). For all three periods, we clearly find an optimum field strength, where the fluctuations within the poling patterns are minimized. However, the duty cycle of the realized poling patterns is not always close to the ideal 50%, which we denote by the filling of the symbols corresponding to the different measurements in Fig. 4(a). For a realized duty cycle between 40% and 60%, we use filled symbols; however, for other duty cycles, they are not filled.

To understand the dependence of total poling inhomogeneity on the poling field strength better, we analyze the experimental result for a period of $\Lambda = 2$ μm in more detail. For poling field strengths of $U_p = 46, 50, 56$ V/μm, we show the realized duty cycle η in dependence on the spatial coordinate z in Fig. 4(b). Corresponding binary images of the obtained domain patterns are shown in Figs. 4(c)–4(e). For $U_p = 46$ V/μm, Fig. 4(c) shows that the growth of domains to reach the negative electrodes is not completed and hence our edge detection method does not find reliable edge positions for the poled domains.

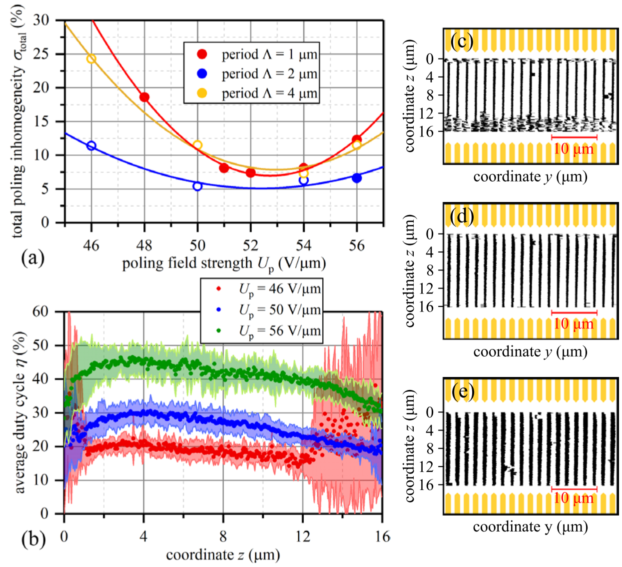

Fig. 4. (a) Total poling inhomogeneity σ_{total} in dependence on the poling field strength U_p for different periods Λ. The filled symbols mark experiments with total average duty cycle η_{total} between 40% and 60%. (b) Variation of the average duty cycle η along the z direction for experiments from (a) with period of $\Lambda = 2$ µm. Shading is the poling inhomogeneity σ for each z. (c)–(e) Poled domains obtained from PCM images for experiments from (a) with a period of 2 µm and poling field strength U_p of (c) 46, (d) 50, and (e) 56 V/µm, respectively. Note that in (b), for U_p of 46 V/µm, the region with $z > 12$ µm shows a large value of poling inhomogeneity, which indicates that the poling is not completed and random values are detected [see (c)]. The same effect can be seen for U_p of 50 V/µm close to $z = 16$ µm.

This results in a sudden jump of the calculated η and larger poling inhomogeneity σ (red shading) for positions corresponding to the unpoled areas at $z \geq 12$ µm [red dots in Fig. 4(b)], which leads to an increased σ_{total} overall. However, in the region corresponding to 2 µm $\leq z \leq 12$ µm, uniform domains are realized, albeit with a reduced average duty cycle of $\eta \approx 20\%$.

Increasing the poling field strength U_p leads to an extension of the domains toward the negative electrode and an increase of the duty cycle. For $U_p = 50$ V/µm [Fig. 4(d)], the domains grow enough to cover the whole 16 µm wide area. However, the calculated average duty cycle for this case [blue dots in Fig. 4(b)] indicates that the domains closer to the electrodes, in particular toward the negative electrode, are narrower than in the center of the poling region, which indicates that the domains do not widen uniformly for all z. Nevertheless, the overall homogeneity is improved and σ_{total} has a smaller value compared to the first experiment. A further increase of U_p to 56 V/µm results in further widening of the domains mostly in the middle of the poling region [green dots in Fig. 4(b)], so that η_{total} for this experiment is around 45%. The inhomogeneous widening leads to an increase of σ_{total}. From these results, we conclude that, in general, the optimization of just one parameter of the poling pulse, such as the poling field strength U_p, is not sufficient to obtain a domain pattern with optimal homogeneity and sufficient duty cycle.

We find a similar behavior for poling periods of $\Lambda = 1$ and 4 µm [red and orange data in Fig. 4(a)]. For all poling periods, the poling field strengths U_p realizing the most homogeneous

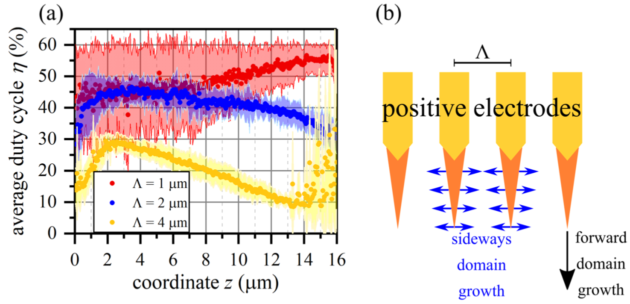

Fig. 5. (a) Spatial dependence of average duty cycle η for different poling periods for a poling field strength of $U_p = 56$ V/µm. For each z, the poling inhomogeneity σ is shown as a shading. For period of $\Lambda = 4$ µm, the domains did not fully develop between positive and negative electrodes, which translated into a large variance of the average duty cycle η for $z > 13$ µm. (b) Scheme of domain growth during poling in forward and sideways directions.

domain patterns are around 52 V/µm. However, for a period of $\Lambda = 1$ µm, η_{total} is within the target range for all experiments, because the narrow domains growing initially are already wide enough and no additional widening is needed to obtain duty cycles of $\approx 50\%$. This is not the case for a period of 4 µm, where wider domains must be realized and all experiments have η_{total} less than 40%.

To investigate the shape of the domains in more detail, the average duty cycle η along the z directions for experiments with $U_p = 56$ V/µm for all three periods is shown in Fig. 5(a). For a period of 1 µm, the domain width is around $\eta = 50\%$ for smaller z; however, it increases for z close to the negative electrode. The reason could be that close to the positive electrodes (smaller z) the sideway growth of the domains is inhibited, either by interaction between neighboring domains or by the influence of the electrodes, which do not allow the domains to grow wider than the electrode width. However, further away from the electrodes, more sideway evolution of the domains seems possible. Besides, the domains are more inhomogeneous for smaller z (closer to positive electrodes), which is shown as a wider shading in the red graph in Fig. 5(a). We assume this inhomogeneity is because the optimal value of U_p for this poling period is smaller than 56 V/µm. For the case of $U_p = 52$ V/µm, we observed that the poling is much more homogeneous, corresponding to the results shown earlier.

The final shape of domains is different for larger poling periods. For a period of 2 µm, the domains are quite homogeneous for $z < 12$ µm, before getting about 10% narrower close to the negative electrode. Finally, for a period of 4 µm, the average domain profile is triangular. The different domain shapes for different periods can be explained by the growth behavior of the domains, which we sketch in Fig. 5(b). The domains grow in two directions: forward toward the negative electrode and sideways, perpendicular to the direction of the field between the electrodes. It has been shown that the local electric field at the domain walls, which is required for domain growth, is affected by the neighboring domain growth. This effect is related to the distance between the adjacent domains, which actually determines the field distribution [36]. Besides that, when two growing domains approach each other, the domain growth slows down [39]. Based on these considerations, we can claim that the

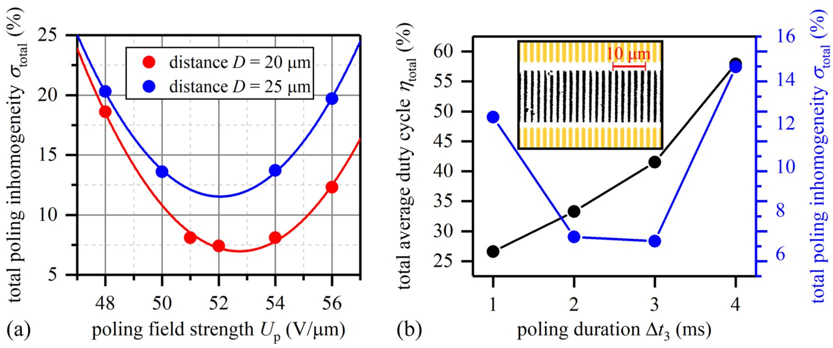

Fig. 6. (a) Dependence of total poling inhomogeneity σ_{total} on the poling field strength U_{p} for different distances between electrodes D. For all experiments, the total average duty cycle η_{total} is between 40% and 60%. (b) Effect of changing Δt_3 on η_{total} and σ_{total}. The inset shows a polarization contrast microscopy (PCM) image of the poled region for the experiment with Δt_3 of 3 ms.

different domain shapes for different poling periods are related to different speeds of sideways domain growth. For a period of 1 μm, the adjacent domains are closer to each other, so the sideways domain growth is slower and domains grow mostly in a forward direction. Hence, the final shape has a quite uniform duty cycle without merging. However, for a period of 4 μm, the neighboring domains are further apart from each other and the domains grow faster in the sideways direction, which results in a triangular domain profile.

In the next set of experiments, we investigate the influence of the distance between electrodes D on the poling quality for the smallest poling period, $\Lambda = 1$ μm. Increasing the distance between the electrodes would enable the fabrication of a larger number of optical elements like waveguides, couplers, and photonic crystals in the area realized with one poling step. To explore this possibility, we analyze poling results for different poling field strengths as before, but for a distance between the electrodes of $D = 25$ μm. We compared the results with the previous experiments. Figure 6(a) shows the variation of σ_{total} for a changing U_{p}. Similar to the previous experiments, we exclude the area close to the electrodes due to low poling quality and a 21 μm wide region of interest is chosen for subsequent analysis. As we can see, the poling quality for experiments with a shorter electrode distance is better. Larger distances between the electrodes cause more domain mergings and inhomogeneities. Hence, increasing the distance decreases the quality of poling.

Finally, we study the effect of the poling duration Δt_3 on the poling quality, to explore whether using a second experimental parameter for optimization enables poling with similarly high homogeneity and a duty cycle close to 50%. To this end, we perform these experiments for a period of $\Lambda = 2$ μm, where the variation of the poling field strength seemed insufficient for realizing an optimal poling pattern. We use a fixed poling field strength of $U_{\mathrm{p}} = 54$ V/μm and vary the poling time Δt_3 from 1 to 4 ms. The measured total average duty cycle η_{total} and total poling inhomogeneity σ_{total} for four experiments is shown in Fig. 6(b). For the case of $\Delta t_3 = 1$ms, the domain's forward growth is not completed and the calculated σ_{total} is high. Increasing Δt_3 to 2 ms completes the forward growth of domains from positive to negative electrodes and decreases the value of σ_{total}; however, the domains are not wide enough with $\eta_{\mathrm{total}} \approx 33\%$. Increasing Δt_3 to 3 ms results in wider domains with $\eta_{\mathrm{total}} \approx 42\%$ with the same total poling inhomogeneity σ_{total} as for a 2 ms poling duration. This shows that the domains widen homogeneously and indicates that controlling the poling duration is a useful technique to widen the domains to reach the desired duty cycle. The poling result from the experiment with $\Delta t_3 = 3$ ms is shown in the inset of Fig. 6(b). A further increase of Δt_3 to 4 ms results in a further widening and partial merging of the domains, which results in a high σ_{total}.

Our results show that the poling duration influences both the forward and sideways growth of the domains, like the poling field strength studied before. However, the dependence of the homogeneity and the duty cycle of the realized poling pattern on the poling duration is significantly different from the impact of the poling field strength. Hence, optimizing both parameters will lead to optimal poling conditions.

Based on our results, we suggest a strategy where the poling field strength U_{p} is adjusted first to complete the forward domain growth from the positive to the negative electrode without inhomogeneously widening the domains. Then the poling duration Δt_3 is optimized to widen the domains and obtain the desired duty cycle.

4. CONCLUSION

In conclusion, we carried out a systematic study to analyze the effect of different experimental parameters on the surface electric field poling (EFP) of X-cut thin-film lithium niobate on insulator (LNOI) with poling periods of 1–4 μm. We investigated the realized poled domains with polarization contrast microscopy and developed an image processing method that enabled a statistical analysis of the poling quality. We showed that applying pre-pulses with weaker field strength improves the final poling quality. Our experiments show the effect of poling field strength and poling duration on the domain growth in the forward and sideways direction. We demonstrated that an optimal poling result can be reached for adapted values for both parameters. The optimal parameters are different for different poling periods and distances between the electrodes, where a larger electrode distance generally decreases the poling quality. Finally, we suggested a procedure to optimize the two main parameters of the poling process.

We believe these results can be used to find the best poling signal for the fabrication of high-quality poled regions large enough to accommodate several integrated optical elements in the same poled region.

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